

Multimodal well-being assessment app

Literature survey

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Mental health challenges such as stress, burnout, and depression are rising globally. A core difficulty is that many people struggle to recognize their own emotional decline until it has escalated, and common assessments rely on infrequent, subjective self-reports that can be biased.

You will design and build a comprehensive mobile application that serves as a mental health monitoring tool. The app will use AI to analyze three key data streams — facial expressions, voice patterns, and responses to structured psychological questionnaires — to assess a user's state of mind. It will track changes over time to detect early warning signs of burnout, depression, suicidal ideation, and other mental health conditions. The system will integrate data from three distinct sources to build a holistic picture of the user's mental state: facial expression analysis, voice analysis, and psychological questionnaire responses. The result is a functional mobile application that lets a user record a short video or audio clip, answer a few questions, and receive a wellness score along with a summary of potential risks.

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1.0 Introduction

Mental health challenges such as stress, burnout, and depression have increased significantly over the years (Biomedical and Pharmacology Journal, 2025). Many individuals fail to recognize their own emotional decline until it has significantly escalated. Those who are gradually burning out or becoming depressed often keep going to work or school without being noticed. By the time a person asks for help, or a loved one notices their condition, it may have already reached a severe stage. This delay in recognition is the core reason early, low-cost, and easy-to-access screening tools are needed (Frontiers in Psychiatry, 2022), especially a tool that a person can use privately and daily, such as a mobile application on their smartphone.

2.0 Mental Health Assessment: Background & Limitation

2.1 Overview of Mental Health Assessment

Stress, burnout, and depression are interconnected, but they share distinct symptoms, behaviors, and psychological signatures, such as facial expression, sleep and speech patterns, and social activity (Biomedical and Pharmacology, 2025). When chronic stress and burnout are left undetected, they can progress into clinical depression, and in severe cases, into suicidal ideation. This makes early identification an important target for intervention (PMC, 2026).

2.2 Traditional Assessment Methods

To assess mental health conditions, trained professionals have relied on structured and semi-structured interviews. These interviews are supported by self-report instruments such as the Patient Health Questionnaire-9 (PHQ-9) and the Generalized Anxiety Disorder-7 (GAD-7) scale (Kroenke, Spitzer and Williams, 2001). They remain the most widely used screening tools in clinical and digital settings because of their established validity and ease of administration.

2.3 Limitation of Self-report and Infrequent Assessment

Even though PHQ-9 and GAD-7 are widely used, self-report tools still depend on a person's willingness and ability to accurately identify and disclose their symptoms (Neuropsychiatric Disease and Treatment, 2021). This is often affected by stigma, low self-awareness, and the episodic nature of clinical visits. Research on using speech to assess suicide risk notes that clinical interviews and self-report questionnaires depend on a person's willingness to disclose distress. This has driven the search for more objective, passively collected markers of mental state that do not rely solely on self-report.

3.0 Facial Expression Analysis for Emotion and Mental State Detection

3.1 Fundamentals of Facial Expression Recognition and Affective Computing

Facial expression analysis is based on the idea that movements of facial muscles provide externally observable, quantifiable signals of a person's internal emotional state. Depression tends to make people less expressive, smile less, show more sadness, and display a narrower emotional range overall. Computer vision systems can be trained to detect these patterns directly from video (ScienceDirect, 2025).

3.2 Facial Action Coding and Emotional Taxonomies

Most automated facial analysis work builds on two features: coding systems that break facial movement down into discrete action units, and categorical emotion taxonomies (happiness, sadness, anger, fear, surprise, disgust). Together, these give deep learning models the labeled structure they need to learn from image and video data.

3.3 Machine Learning and Deep Learning Approaches to FER

A systematic review of deep learning-based depression recognition from 2017 to 2024 found real progress in the field. One notable finding was that models capturing multiscale spatial-temporal information across video frames perform noticeably better than models working from single static images (ScienceDirect, 2025). CNNs paired with recurrent structures such as LSTMs have become a common way to model both what a face looks like and how it changes over time. One study, for instance, combined LSTM networks with attention over facial action units, head pose, and eye gaze to estimate depression severity from interview footage (MDPI/PMC, 2023). General-purpose systems also reach top scores on benchmarks such as CK+, RAF-DB, FER2013, and FERPlus, which shows how far facial emotion recognition has advanced.

3.4 Evidence Linking Facial Cues to Depression, Stress and Burnout

There is direct clinical evidence tying facial dynamics to depressive states. A frame-by-frame comparison of simulated-depression and normal groups found that the simulated-depression group showed markedly less variation in expressed emotion across a recording, more sadness, and fewer neutral or fearful expressions (PMC, 2024). This lines up with the theoretical link between blunted affect and depression, and it justifies treating expressivity itself, not just discrete emotional labels, as a candidate feature.

3.5 Relevant Dataset

The field draws on two broad categories of dataset:

- General-purpose emotion sets (CK+, FER2013, FERPlus, RAF-DB): used for pretraining and benchmarking.
- Clinically oriented interview corpora (DAIC-WOZ, E-DAIC): pair interview video with validated depression severity scores. These have become the default choice for training and evaluating depression-specific facial models.

3.6 Limitation and Challenges

Facial detection is sensitive to lighting, camera angle, and partial occlusion. People also express emotion differently across cultures and individuals. The bigger structural problem is the scarcity of large, diverse, clinically labeled video datasets. Privacy concerns make it difficult to record and store identifiable video of real patients, which limits how well these models generalize in practice.

4.0 Voice and Speech Analysis for Mental State Detection

4.1 Acoustic and Prosodic Features Relevant to Mental Health

Researchers examine specific voice markers to assess mental health, including pitch, fundamental frequency, jitter, shimmer, speech rate, and pause patterns, as well as glottal and phonation characteristics. These features change when someone feels anxious or depressed, because body movement and vocal control slow down (Journal of Technology in Behavioral Science, 2024).

4.2 Speech Emotion Recognition Techniques

Analyzing acoustic and prosodic speech features with machine learning provides strong clues about mood changes and depression levels. Studies show these audio signals can help track how symptoms change over time, but different studies use different setups, which makes it hard to bring these tools into actual clinical use. Deep learning methods such as convolutional neural networks, GRUs, and BiLSTMs are also used. Recent work uses pretrained speech representation models such as Wav2Vec 2.0 to extract emotional information directly from audio (arXiv, 2025).

4.3 Evidence Linking Vocal Markers to Depression and Suicidal Ideation

More recent work has focused on detecting suicide risk rather than general depression. One study built a classifier from 588 narrative audio recordings submitted through a mobile app by military veterans screening for suicidal ideation, using prosodic, phonation, and glottal characteristics; it showed that smartphone audio works well for this kind of detection (BioData Mining, 2021). Another project used natural language processing on interview text to separate high-suicide-risk cases from non-risk cases (PMC, 2023). Both concluded that the acoustic and linguistic content of speech carries usable signals. Other work has extended into longitudinal tracking of recovery, worsening, and stable depressive states in adolescents in treatment, and separately, used RNN classifiers on voice-journal recordings to distinguish manic from depressive mood states (PMC, 2024).

4.4 Relevant Dataset and Corpora

Voice studies frequently use the DAIC-WOZ interview corpus because it links audio files with PHQ-8 severity scores. Other researchers collect custom datasets, such as veteran phone recordings or clinical interviews, largely because of the scarcity of large, publicly available, clinically annotated speech datasets in this domain.

4.5 Challenges

Background noise creates significant problems for everyday mobile apps. Differences in phone microphones, accents, and language make it hard for a single model to work everywhere. Most research relies on small samples and inconsistent testing protocols, so high accuracy in a lab does not always hold up when tested on new people or new recording setups.

5.0 Psychological Questionnaires and Digital Assessment Tools

5.1 Standardized Instruments

A meta-analytic review found that the PHQ-9 works well on mobile phones, spotting depression about as effectively as a paper test (Neuropsychiatric Disease and Treatment, 2021). This self-report instrument is used widely because it is so well validated. Clinicians and researchers frequently pair it with the GAD-7 to screen for anxiety in both clinical and digital settings.

5.2 Digitalization of Questionnaires in Mobile Health Apps

Tracking mental health symptoms over time works well through smartphone or desktop apps, and this approach fits naturally into measurement-based care. Many mHealth platforms embed the PHQ-9 to spot elevated depressive symptoms across different countries and groups (Frontiers in Psychiatry, 2022). They also combine the PHQ-9 with the GAD-7 to measure depression and anxiety together during remote check-ins.

5.3 Validity and Reliability of Digital Self-Report

Digital self-report tools still face the basic flaws of paper forms: users deal with recall bias or social desirability bias, and many stop filling them out over time. Digitization does bring better privacy and convenience, however, and researchers increasingly blend questionnaire data with passively collected behavioral signals such as facial expression and voice, which works better than relying on a single data stream alone.

6.0 Multimodal Data Fusion for Mental Health Prediction

6.1 Rationale for Combining Multiple Modalities

No single modality is enough to capture something as complex as emotional and psychological state, and relying on one data stream is not sufficient for a robust assessment. Combining behavioral analytics across voice, text, and facial expression with machine learning is consistently reported to improve diagnostic accuracy and reliability over unimodal approaches.

6.2 Fusion Strategies

Researchers take different approaches to combining modalities. Early feature-level fusion merges raw or lightly processed features before classification. Late decision-level fusion trains a separate model per modality and combines their outputs. Hybrid approaches split the difference. One study fused text, audio, and visual data through CNNs for emotion analysis and found it outperformed unimodal approaches (npj Mental Health Research, 2024). Another proposed a memory fusion network to combine verbal, visual, and acoustic streams for mental health prediction (Scientific Reports/PMC, 2025). More recently, attention mechanisms have been used to weight each modality's contribution dynamically, and graph neural networks have been applied to model interactions across voice, text, and facial data, with some reporting accurate results on anxiety and depression detection.

6.3 Existing Multimodal Mental Health Systems

One depression-detection pipeline runs facial expression through a CNN-LSTM, text through an LSTM, and audio through a separate LSTM, then uses transfer learning and ensembling to combine them. Another, working from video, audio, and text in DAIC-WOZ and E-DAIC, combined VGG-16 for speech, BERT for language, and ResNet-50 for vision; the fused model beat a text-only baseline, though only modestly, and text remained the strongest single modality throughout (Scientific Reports/PMC, 2025). A separate framework, the Integrative Multimodal Depression Detection Network, fused video, audio, and text through a Kronecker-product-based mechanism and reported that combining modalities gave a more comprehensive and robust read on depressive symptoms than any single stream alone (PMC, 2025). A fourth system used three dedicated per-modality branches — audio, video, and text — feeding into a shared feature vector, built specifically to move past the limits of purely questionnaire-based screening (PMC, 2024).

6.4 Machine Learning Architectures Used for Fusion

Across all of these systems, CNNs, LSTMs, GRUs, and BiLSTMs remain the default blocks for per-modality feature extraction, with pretrained transformer models such as BERT for text and Wav2Vec 2.0 for audio increasingly taking on that role, and attention-based or graph-based layers handling the actual fusion into a single prediction (arXiv, 2025). Taken together, this body of work supports the technical approach of this project: combining facial, voice, and questionnaire data into one wellness assessment (npj Mental Health Research, 2024).

7.0 Mobile Health (mHealth) Application for Mental Wellness

7.1 Overview of Existing Mental Health Monitoring Apps

Estimates put the number of publicly available mental health apps somewhere between ten and twenty thousand, so the market is already crowded (PLOS Digital Health, 2022). A thematic and sentiment analysis of 92 screening and diagnostic apps from the Apple App Store, Google Play, and the mHealth Index and Navigation Database found depression to be by far the most targeted condition, with most apps screening for a single condition rather than offering anything integrated. A separate scoping review of AI-enabled mental health apps found seventeen studies evaluating apps that used AI for conversational support, mood prediction, and risk assessment, but noted that rigorous outcome evidence for those AI features is still thin (PLOS Digital Health, 2022).

7.2 Comparative Strengths and Gaps

Existing apps do well on accessibility, cost, and privacy relative to in-person care, and PHQ-9-based screening apps have shown good sensitivity for identifying users with high depressive symptoms across different populations (Biomedical and Pharmacology Journal, 2025). Most apps still rely on a single modality, usually questionnaire-based self-report or conversational text, and very few combine facial, vocal, and questionnaire-based signals into one assessment delivered in a single mobile session.

7.3 Design Considerations for AI-Based Mobile Health Application

A few design factors come up repeatedly in the mHealth literature. User engagement matters, because dropout is a major problem for screening and diagnostic apps. There is a real trade-off between on-device and cloud-based inference, weighing responsiveness against privacy, and risk information needs to be presented clearly and without causing alarm, especially where an app might need to flag elevated suicidal ideation risk back to the user.

8.0 Ethical, Privacy and Safety Consideration

8.1 Data Privacy and Security

Facial video, voice recordings, and questionnaire responses are all highly sensitive personal data, so it is not surprising that the literature treats data privacy, secure storage, and informed consent as non-negotiable requirements for any AI-based mental health tool. A systematic review of speech-based suicide risk assessment was blunt about this: robust anonymization and explicit patient consent are prerequisites, not nice-to-haves, and systems also need safeguards against cybersecurity risks such as data breaches before they can be adopted more widely.

8.2 Ethical Concerns in AI-Based Screening

One recurring ethical concern is that AI output could be mistaken for a clinical diagnosis rather than a screening signal (The Hastings Center, 2024). Researchers building AI models for mental health content classification have responded by explicitly framing their systems as screening assistants for human review, not diagnostic tools, particularly for high-signal categories such as suicide-related content. The literature also raises algorithmic bias, the difficulty of interpreting black-box models in a clinical context, and the risk that ordinary situational stress could be misread as pathological if social context is ignored (Taylor & Francis, 2026).

8.3 Responsible Handling of Suicidal Ideation Risk

Ethical guidance for AI-based suicide risk tools tends to converge on a small set of principles: collect only the minimum data necessary, explain clearly and accessibly how data will be used and stored, use strong encryption and access controls, keep a human in the loop for any escalation decision, and leave final clinical judgment with qualified professionals rather than the algorithm. Ethical analyses of AI in suicide prevention also emphasize informed consent, respect for autonomy and dignity, and transparency about the existence and purpose of algorithmic risk screening, since users may not expect their data to be evaluated for anything beyond their immediate reason for using the app (The Hastings Center, 2024; Taylor & Francis, 2026).

9.0 Research Gap and Summary

9.1 Synthesis of Findings

Three points come through clearly across this literature. Facial expression analysis and voice analysis have each been validated independently as viable and reasonably objective sources of information about depressive and anxious states, backed by a growing body of deep learning research and clinically labeled datasets such as DAIC-WOZ. Standardized questionnaires such as the PHQ-9 remain reliable and well validated, and moving them onto a mobile interface does not appear to cost them any diagnostic value. Multimodal fusion of facial, vocal, and textual or questionnaire data consistently outperforms any single modality used alone, which is the technical premise this project is built on.

9.2 Identified Gaps

Most multimodal research is still conducted on offline datasets such as DAIC-WOZ rather than deployed as a real-time, mobile-first application. Most commercially available mental health apps still rely on a single modality, usually questionnaire-based self-report or a text chatbot, rather than combining facial, vocal, and questionnaire data. Facial and voice datasets remain limited in size and diversity because of legitimate privacy constraints, which caps how generalizable any model built on them can be and makes careful, privacy-conscious data handling a requirement rather than an afterthought for any new system (ScienceDirect, 2025). While ethical guidance for AI-based suicide risk

screening is well developed at the level of principle, few applied systems in this literature show how those principles — such as minimal data collection and transparent consent — get operationalized end-to-end inside a single consumer-facing app.

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